

The 2-Kazhdan question for integral special linear groups

Literature-implied answer note for arXiv:1711.10238

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Status. Literature-implied answer, complete for $n \geq 3$. The large- n clause is affirmative: $\mathrm{SL}_n(\mathbb{Z})$ is 2-Kazhdan for every $n \geq 4$. The case $n = 3$ is negative.

Source question

Question 4.6 of De Chiffre–Glebsky–Lubotzky–Thom [1], on source PDF page 24, asks:

Is $\mathrm{SL}_n(\mathbb{Z})$ 2-Kazhdan (at least for large n)?

In the source terminology this means

$$H^2(\mathrm{SL}_n(\mathbb{Z}), V) = 0 \quad \text{for every unitary representation } V.$$

Identification

Bader–Sauer’s Theorem A [2], on supporting PDF page 2, states that $\mathrm{SL}_n(\mathbb{Z})$, as a lattice in $\mathrm{SL}_n(\mathbb{R})$, has property (T_{n-2}) . Thus, when $n \geq 4$,

$$H^2(\mathrm{SL}_n(\mathbb{Z}), W) = 0 \quad \text{whenever } W^{\mathrm{SL}_n(\mathbb{Z})} = 0.$$

Their displayed formula (1) on the same page shows $H_c^2(\mathrm{SL}_n(\mathbb{R}), \mathbb{C}) = 0$. Their Theorem C transfers continuous cohomology to the lattice in degrees strictly below the real rank $n - 1$, so

$$H^2(\mathrm{SL}_n(\mathbb{Z}), \mathbb{C}) = 0 \quad (n \geq 4).$$

Now split an arbitrary unitary representation orthogonally as $V = V^\Gamma \oplus (V^\Gamma)^\perp$, where $\Gamma = \mathrm{SL}_n(\mathbb{Z})$. The second summand has zero degree-two cohomology by property (T_{n-2}) . The action on V^Γ is trivial; because Γ is of finite type in degree two, its finite cochain complex gives

$$H^2(\Gamma, V^\Gamma) \cong H^2(\Gamma, \mathbb{C}) \otimes V^\Gamma = 0.$$

Hence $\mathrm{SL}_n(\mathbb{Z})$ is 2-Kazhdan for every $n \geq 4$.

The endpoint below this range is determined by Brück–Hughes–Kielak–Mizerka’s Theorem 1.1 [3], on supporting PDF page 2. They construct a finite-dimensional orthogonal representation π_3 such that

$$H^2(\mathrm{SL}_3(\mathbb{Z}), \pi_3) \neq 0.$$

Therefore $\mathrm{SL}_3(\mathbb{Z})$ is not 2-Kazhdan.

Conclusion, provenance, and scope

Combining the two later papers gives the sharp classification for $n \geq 3$:

$\mathrm{SL}_3(\mathbb{Z})$ is not 2-Kazhdan, while $\mathrm{SL}_n(\mathbb{Z})$ is 2-Kazhdan for every $n \geq 4$.

This is an agent-identified implication of known theorems, not new mathematics. The supporting papers work in the source’s higher-property-T framework, but neither presents this exact classification as an answer to Question 4.6. Exact-question, exact-title, and core-keyword searches through 11 August 2026 led to these decisive sources.

This note does not address the source’s separate questions about amenable or solvable Frobenius approximability, central quotients, semidirect products by $\mathbb{Z}$, or cohomology with arbitrary C^* -algebra coefficients.

References

- [1] M. De Chiffre, L. Glebsky, A. Lubotzky, and A. Thom, *Stability, cohomology vanishing, and non-approximable groups*, Forum Math. Sigma 8 (2020), e18; arXiv:1711.10238.
- [2] U. Bader and R. Sauer, *Higher Kazhdan property and unitary cohomology of arithmetic groups*, arXiv:2308.06517v3 (2026). See Theorem A, Theorem C, and formula (1).
- [3] B. Brück, S. Hughes, D. Kielak, and P. Mizerka, *Non-vanishing unitary cohomology of low-rank integral special linear groups*, Int. Math. Res. Not. 2025, no. 15, rnaf230; arXiv:2410.22310. See Theorem 1.1.